



Facilitating the Integration of Borehole Thermal Energy Storage in District Energy Systems: Improving the Fidelity of Design Tools

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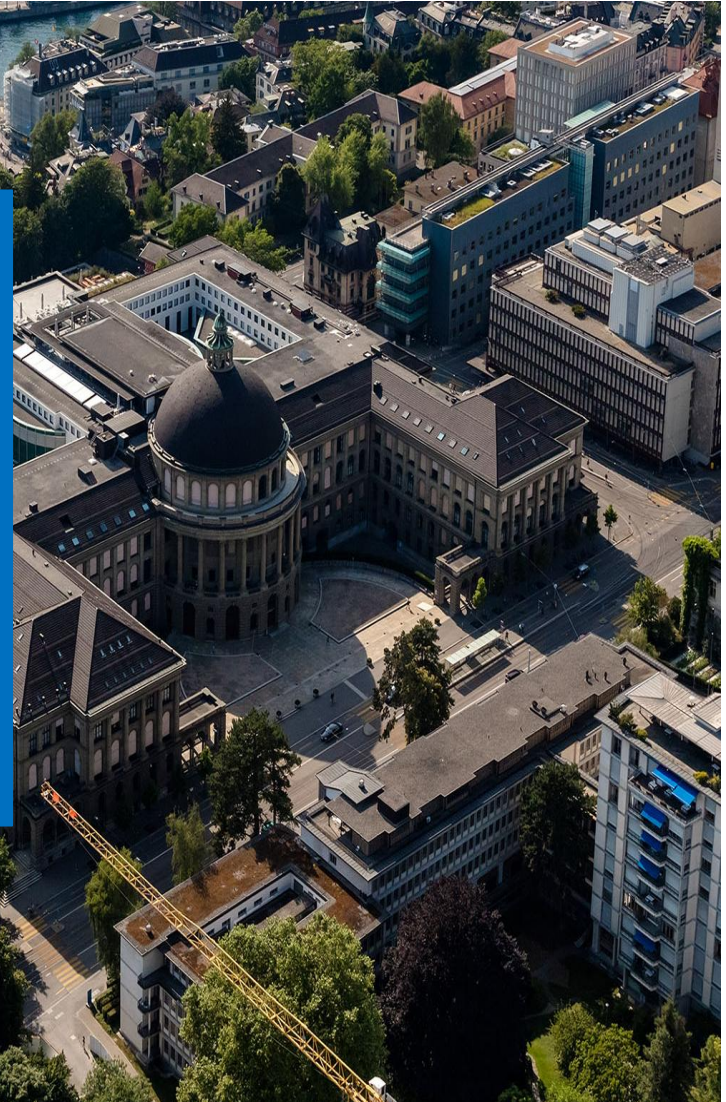
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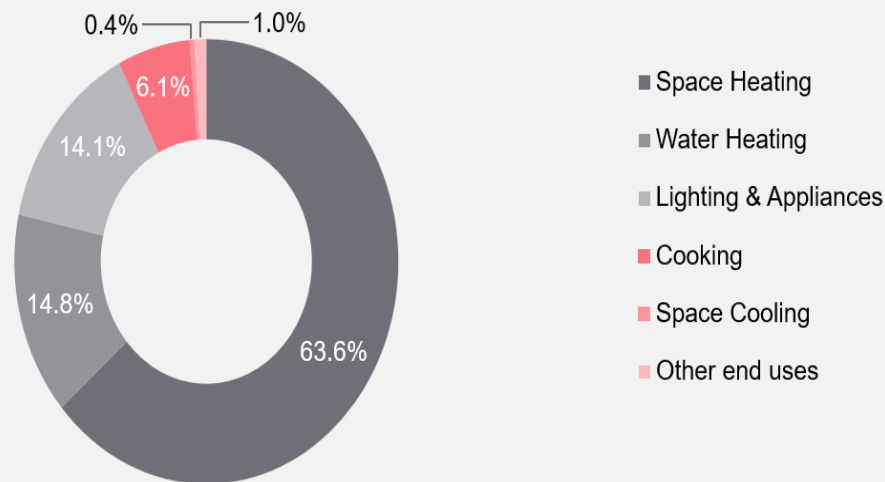
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- In the EU 49.3% of net electricity generated comes from renewable energies¹
- For heating and cooling only 26.1%¹



Final Energy Consumption in the residential Sector by End-use, 2018



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[1] https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Renewable_energy_statistics

[2] <https://www.futurebridge.com/industry/perspectives-energy/renewable-based-heating-and-cooling-in-europe-by-2030/>

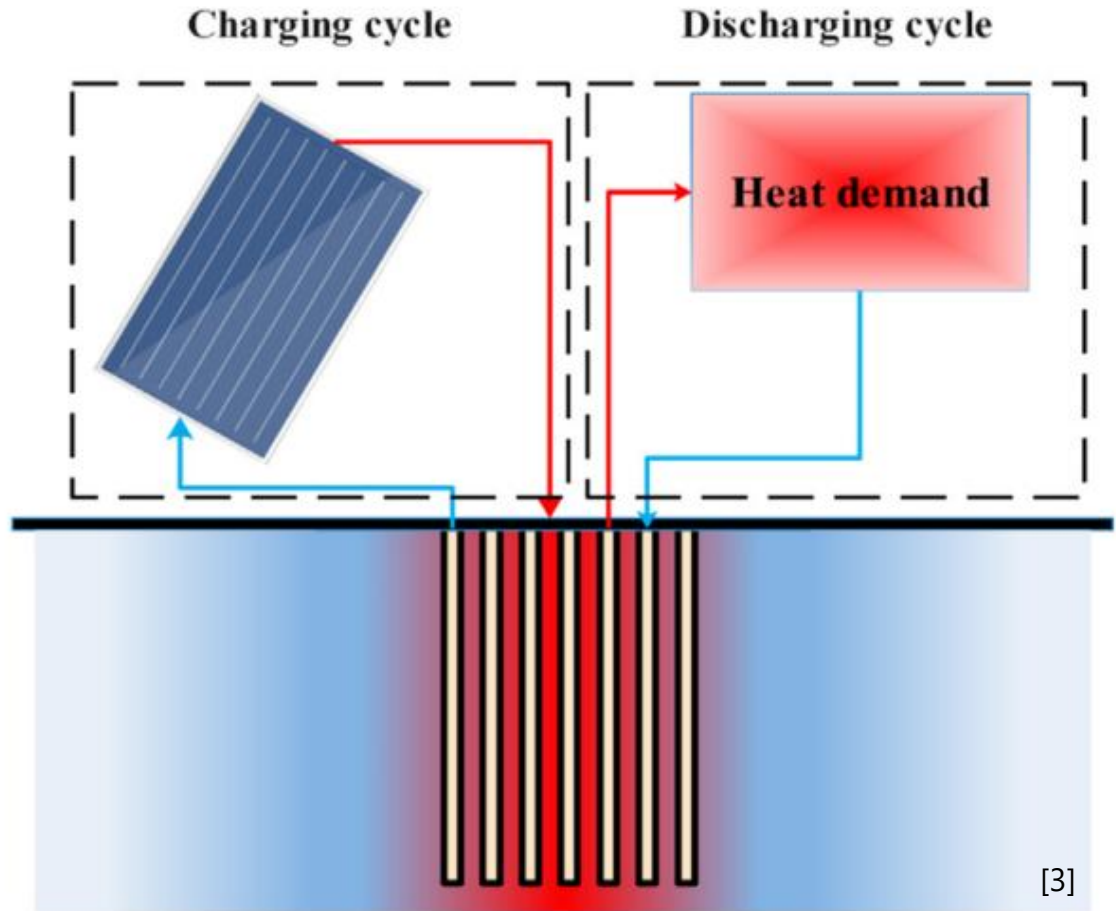
Why is Borehole Thermal Energy Storage (BTES)?

BTES to contribute to the transition⁴

- Utilize the ground as a thermal storage medium by injecting and extracting heat through the borehole heat exchangers⁵
- Enable seasonal thermal energy storage, allowing surplus heat generated during one period of the year to be utilized at a later time⁶

BTES limitations

- High initial investment
- Large land requirement
- Site-dependent performance



[3] Pourahmadiyan, A., Sadi, M., & Arabkoohsar, A. (2023). *Seasonal Thermal Energy Storage*. In A. Arabkoohsar (Ed.), *Future Grid-Scale Energy Storage Solutions: Mechanical and Chemical Technologies and Principles* (pp. 215–257)

[4] Mohammadamin Ahmadfard and Ehsan Baniasadi. Borehole thermal energy storage systems: A comprehensive review using bibliometric and qualitative tools.

[5] <https://dewesoft.com/blog/solar-borehole-thermal-energy-storage>

[6] Cynthia Ann Cruickshank and Christopher Baldwin. Sensible thermal energy storage: Diurnal and seasonal. In *Storing Energy*, pages 291–311. Elsevier.

Modern BTES Research

- ☐ Overview of BTES operation⁷
- ☐ MES sizing and operation optimization⁸

Existing literature does not clearly quantify:

- ☐ Which parameters most affect BTES model fidelity
- ☐ Whether simplified models remain suitable for system-level optimization

Trade-Off

- ☐ **High-Fidelity Models, e.g. Modelica Models^{9,10}**
 - Used for operation modeling
 - Better thermal representation
 - Very high computational cost
- ☐ **Simplified Models, e.g. MINLP, MILP¹¹**
 - Sizing tools
 - Faster optimization
 - Reduced physical realism

[7] Manfred Reuss. The use of borehole thermal energy storage systems. In Advances in Thermal Energy Storage Systems.

[8] Pierluigi Mancarella. MES (multi-energy systems): An overview of concepts and evaluation models

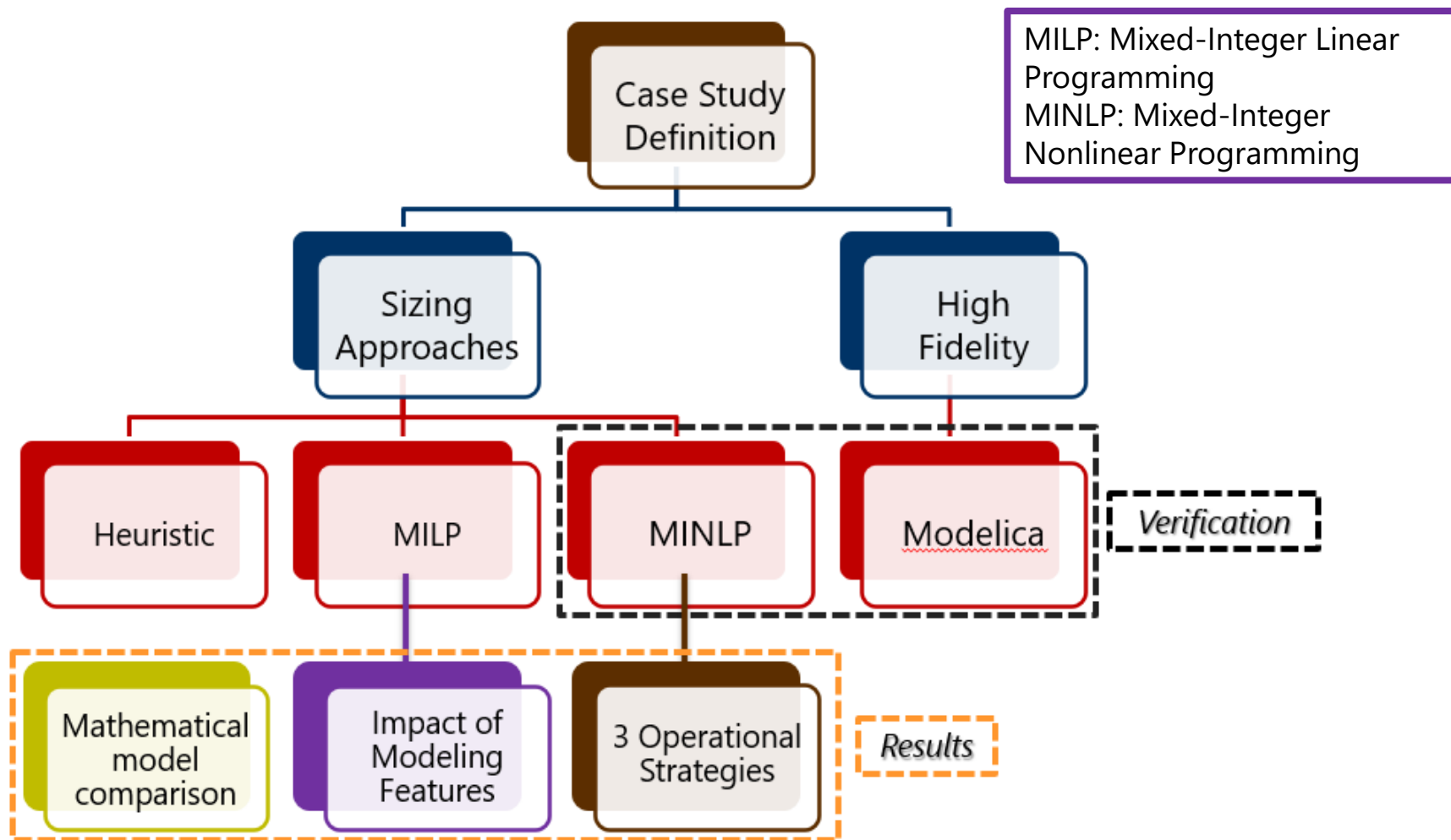
[9] Mohamed T. Bahr, Piyush Kumar Kumawat, Palash Panja, and Kody M. Powell. A dynamic 2d borehole thermal energy storage (BTES) model for enhanced computational efficiency. 4

[10] Mohamed T. Bahr, Ali Khosravani, Dynamics simulation and economic evaluation of dual-mode borehole thermal energy storage in electrified district energy systems.

[11] Massimo Fiorentini, Philipp Heer, and Luca Baldini. Design optimization of a district heating and cooling system with a borehole seasonal thermal energy storage.

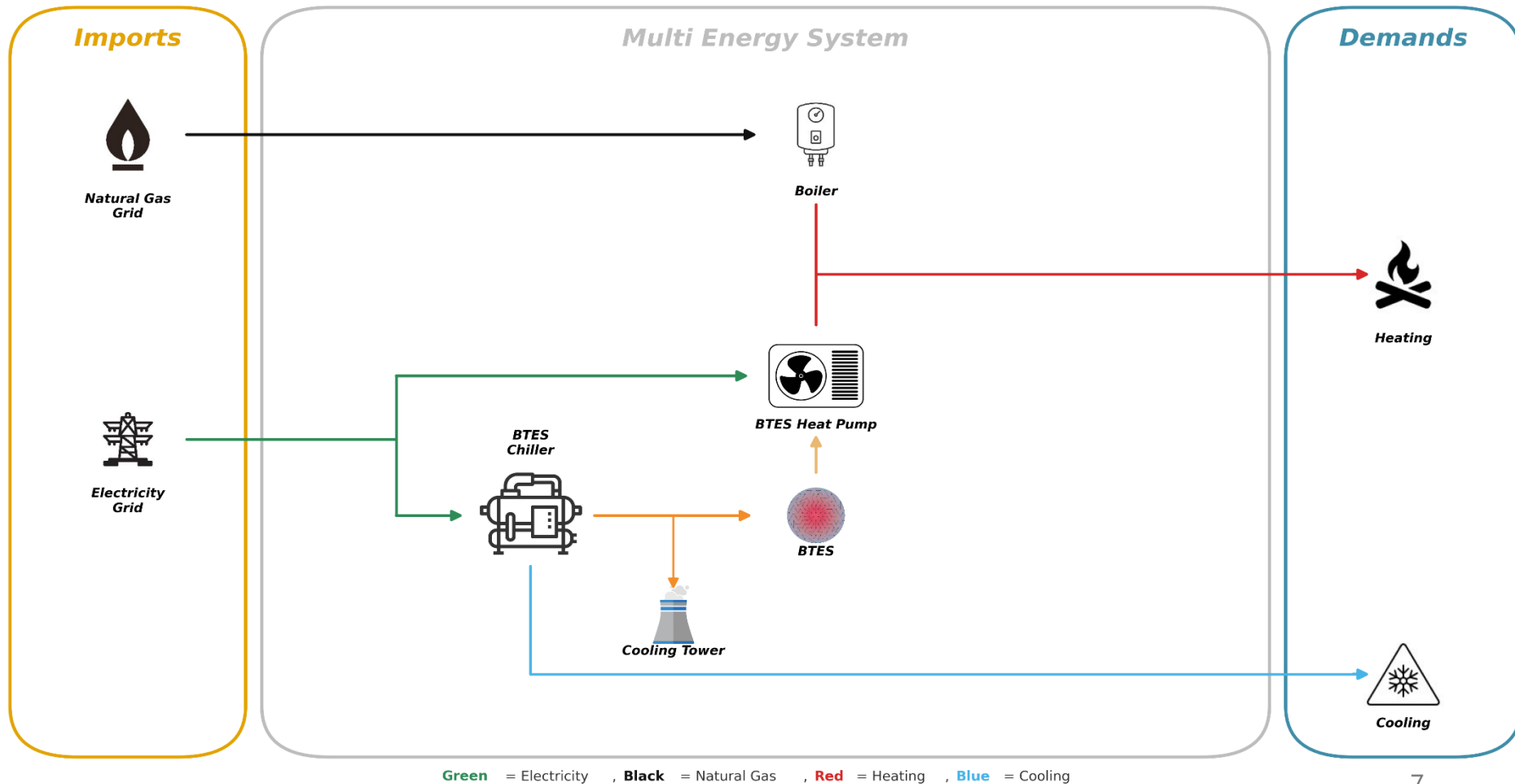
How different modeling assumptions influence the optimal design, operation, and computational performance of BTES-integrated energy systems?

1. Quantify how different operational assumptions influence technology sizing, system operation, annual system cost and computational performance?
2. How does the level of detail of the mathematical formulation affect the results?
3. Is the level of system complexity an important aspect of model selection?

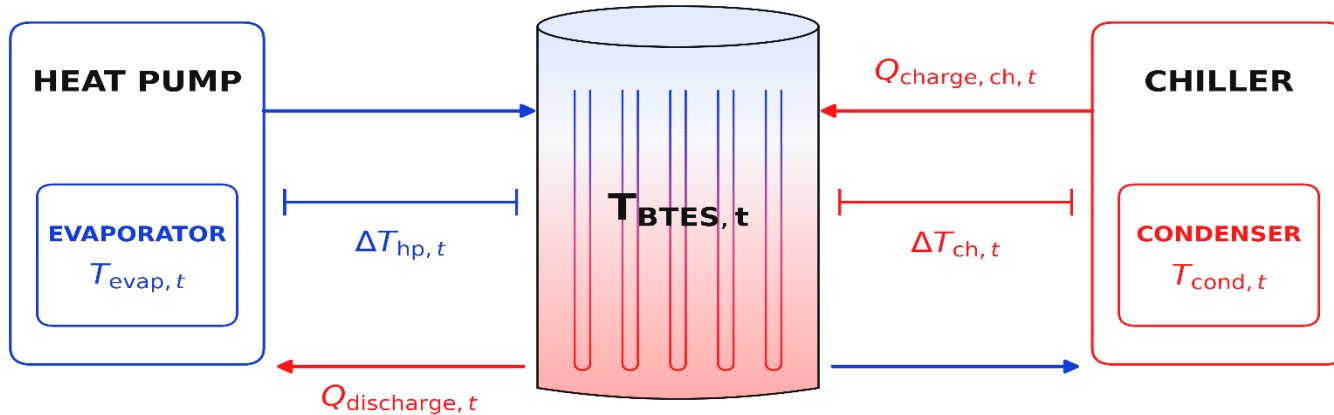


Methodology - Case Study Definition

■ EMPA campus case study



Borehole Thermal Energy Storage



Note: All strategies tested using the MINLP model

$$\frac{1}{COP_{HP}} = \alpha(T_{btes}[t] - \Delta T_{hp,btes}[t]) + \beta$$

$$\frac{1}{COP_{chiller}} = \alpha(T_{btes}[t] + \Delta T_{chiller,btes}[t]) + \beta$$

Strategy I	Strategy II	Strategy III
ΔT as scalar decision variable	T_{evap}/T_{cond} as scalar decision variables	$\Delta T/T_{evap}/T_{cond}$ as varying decision variables

Methodology – McCormick relaxation - MILP

Bilinearity:

$$P = Q \times \frac{1}{COP}$$

Lower bounds:

$$z \geq (Q_{LB} \times T) + (Q \times T_{LB}) - (Q_{LB} \times T_{LB})$$

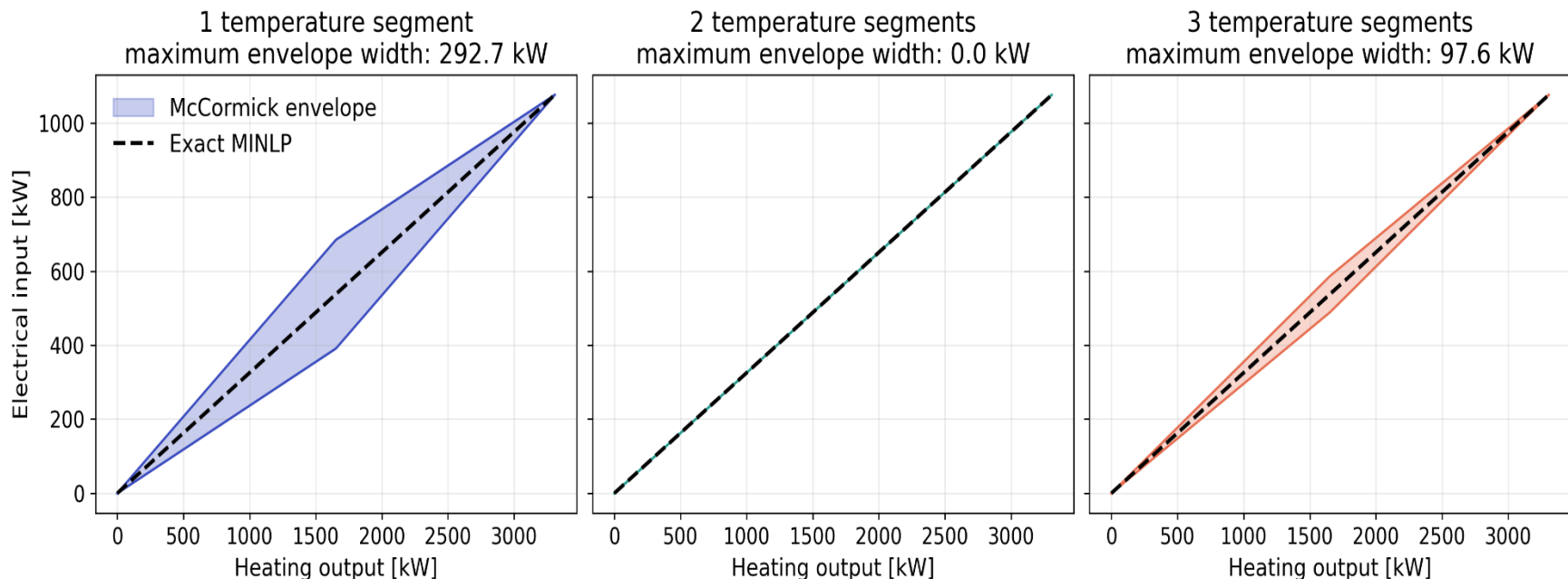
$$z \geq (Q_{UB} \times T) + (Q \times T_{UB}) - (Q_{UB} \times T_{UB})$$

Upper bounds:

$$z \leq (Q_{UB} \times T) + (Q \times T_{LB}) - (Q_{UB} \times T_{LB})$$

$$z \leq (Q_{LB} \times T) + (Q \times T_{UB}) - (Q_{LB} \times T_{UB})$$

McCormick approximation of the heat-pump bilinear term at $T_{evap} = 10^\circ \text{C}$



Option	Diameter (m)	Depth (m)	Volume (m ³)	UA (kW/K)	Boreholes
1	53.4	53.4	119,200	22.5	158
2	61.2	61.2	179,600	33.8	207
3	70.0	70.0	269,400	50.6	271
4	80.1	80.1	404,100	76.0	355
5	91.7	91.7	606,100	113.9	466
6	—	—	—	—	—

Results - Verification

Heat-exchanger temperature comparison: condenser during charging, evaporator during discharging



Good agreement with Modelica (MAE = 2.51 K, RMSE = 2.84 K),
supporting the MINLP for annual system-design optimization

Results – Impact of Operational Strategies

Technology	Strategy I (Scalar ΔT)	Strategy II (Scalar Tevap/Tcond)	Strategy III (Varying ΔT /Tevap/Tcond)
Heat Pump (kW)	1019	1139	1227
Cooling Tower (kW)	810	723	610
BTES	4	4	4
Volume (m ³)	404,100	404,100	404,100
Annualized Cost (kCHF/a)	987	995	968
Run time (s)	42.99	42.63	37.11

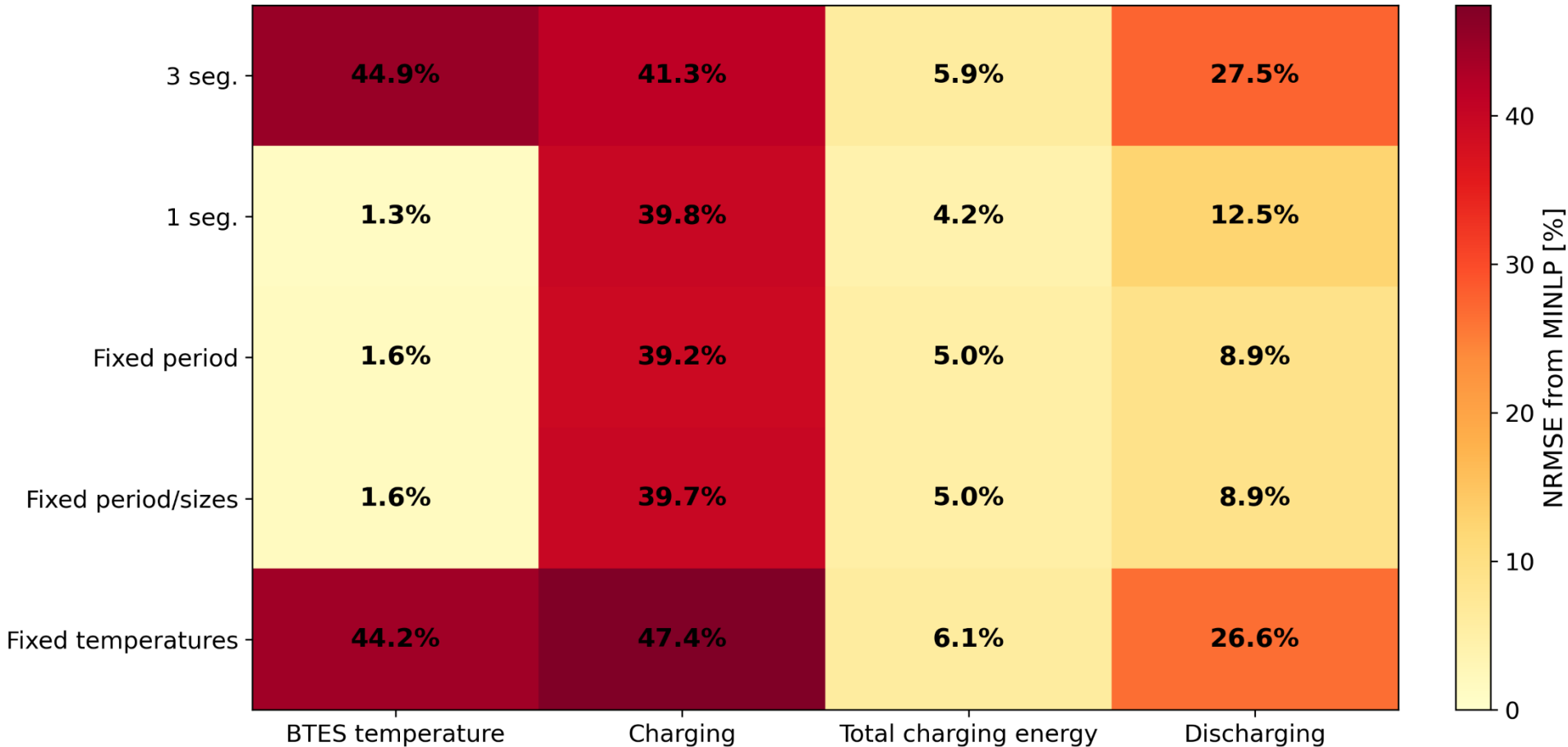
Increasing operational flexibility changes technology sizing and reduces the annualized system cost by approximately **2.7%**, while all strategies select the same BTES configuration

Results - MILP simplifications/sizes

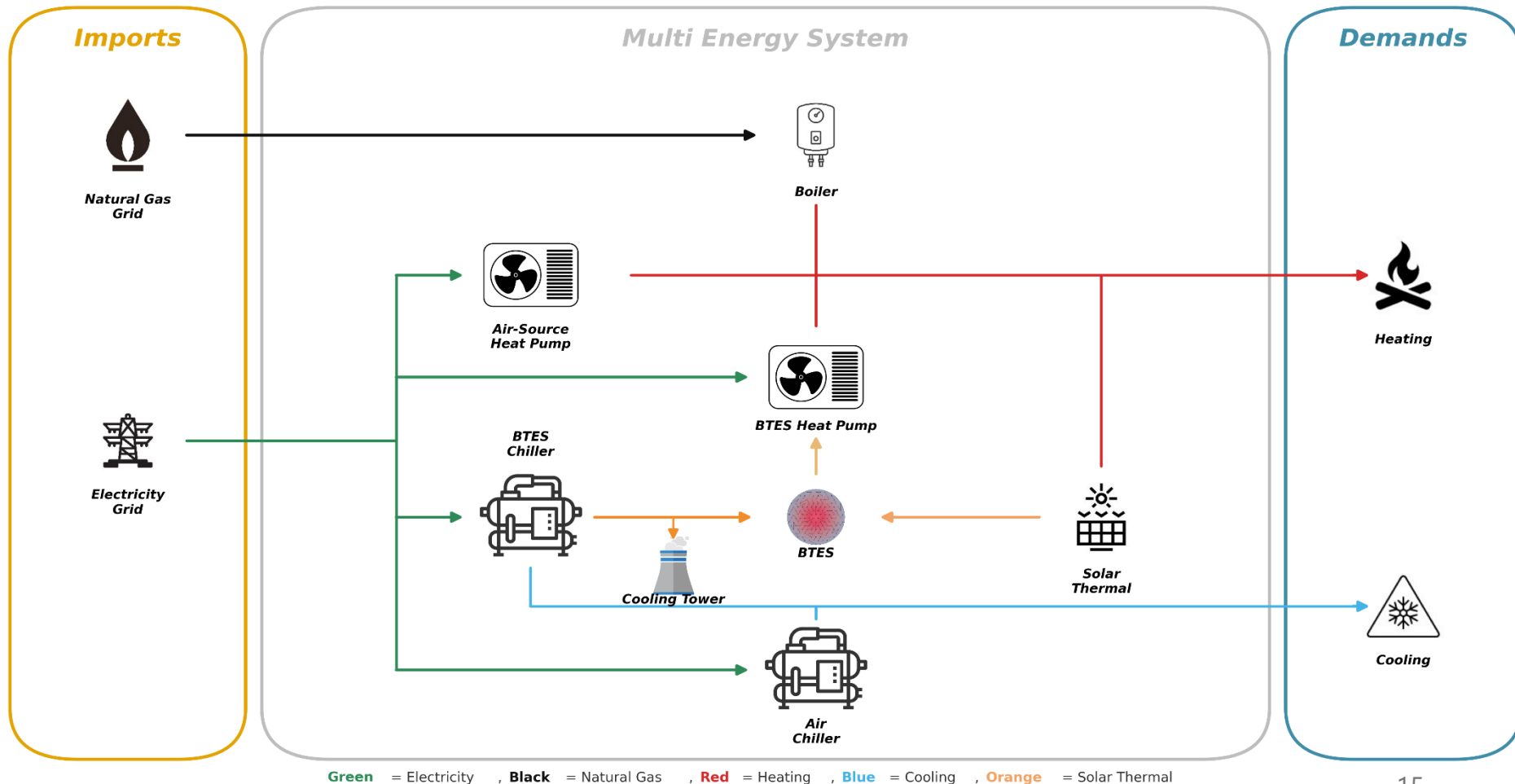
Quantity	MINLP	3 seg.	1 seg.	Fixed period	Fixed period/sizes	Fixed temp.
Speedup relative to MINLP	1.00	0.41	0.70	1.32	1.11	10.67
Cost difference from MINLP	0.00	6.11	-1.15	-0.88	-0.88	6.23
Temperature RMSE [K]	0.000	7.064	0.200	0.248	0.252	6.957

Results - MILP simplifications/operation

Operational differences relative to the EMPA MINLP



■ General case study



- 1 segment fixed period MILP used
- 1.45 speed foldup compared to the MINLP
- Higher foldup than the simple case study

Increased system
flexibility made the
linearization even more
attractive

Operational Strategies

- Increasing operational flexibility reduces the annual system cost by approximately 2.7%

Mathematical Formulation

- The 3-segment MILP increased computational effort without improving agreement with the MINLP
- The 1-segment MILP consistently reproduced the MINLP more closely than the original 3-segment MILP

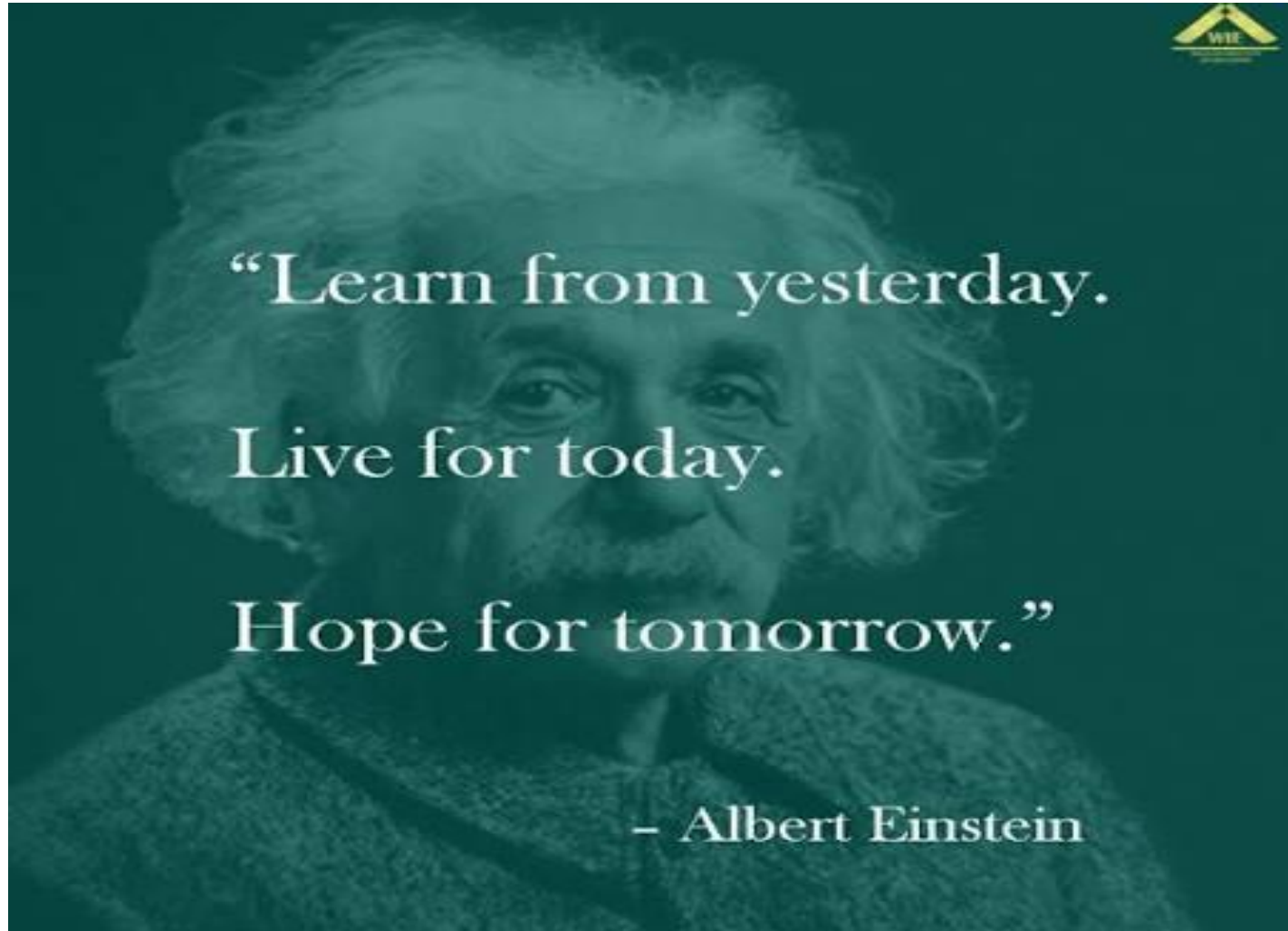
Modelling simplifications

- The 1-segment fixed-period MILP provides the best compromise between accuracy and tractability
- Fixing the operating temperatures provides the largest speed-up but also the largest deviation from the MINLP

System Complexity

- The computational advantage of the simplified MILP increases as system flexibility increases

Quote of the Day



Appendix

Lower bounds:

- $z \geq (Q_{LB} \times T) + (Q \times T_{LB}) - (Q_{LB} \times T_{LB})$
- $z \geq (Q_{UB} \times T) + (Q \times T_{UB}) - (Q_{UB} \times T_{UB})$

Upper bounds:

- $z \leq (Q_{UB} \times T) + (Q \times T_{LB}) - (Q_{UB} \times T_{LB})$
- $z \leq (Q_{LB} \times T) + (Q \times T_{UB}) - (Q_{LB} \times T_{UB})$

Bilinear relation

$$z_{hp,t,s} = Q_{hp,t,s} T_{evap,t,s}$$

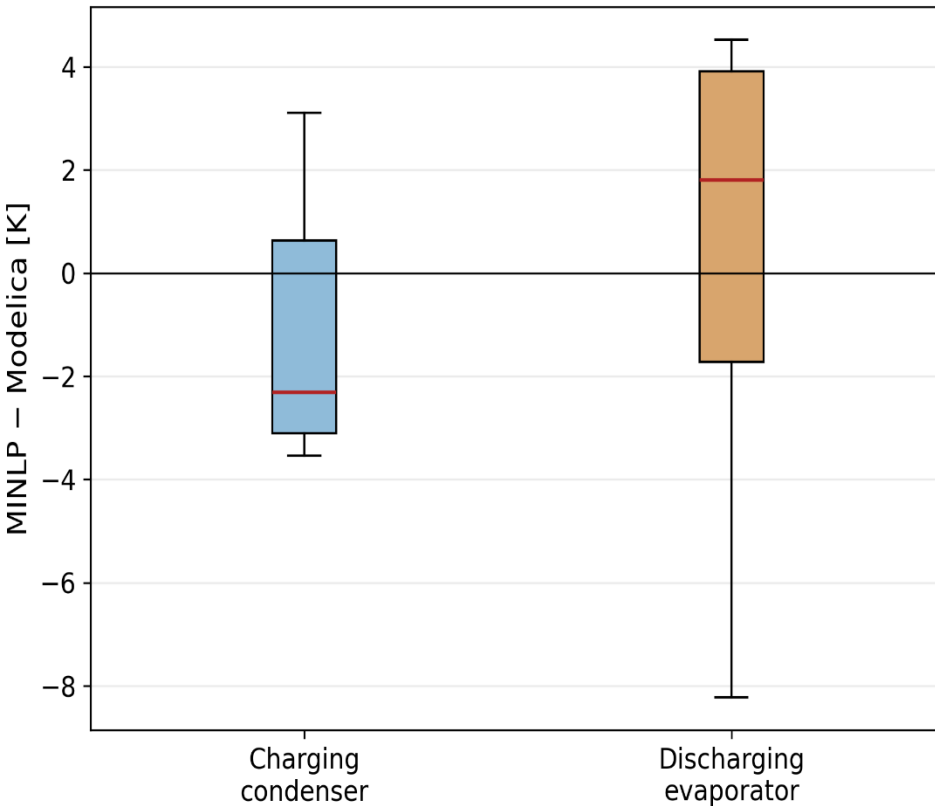
Segment activation and variable bounds

$$\begin{aligned} \sum_{s \in \mathcal{S}} y_{hp,t,s} &= 1, & y_{hp,t,s} &\in \{0,1\}, & \forall t. \\ 0 &\leq Q_{hp,t,s} \leq \bar{Q}_{hp,t} y_{hp,t,s}, & \forall t, s. \\ T_{evap,s}^{L} y_{hp,t,s} &\leq T_{evap,t,s} \leq T_{evap,s}^{U} y_{hp,t,s}, & \forall t, s. \end{aligned}$$

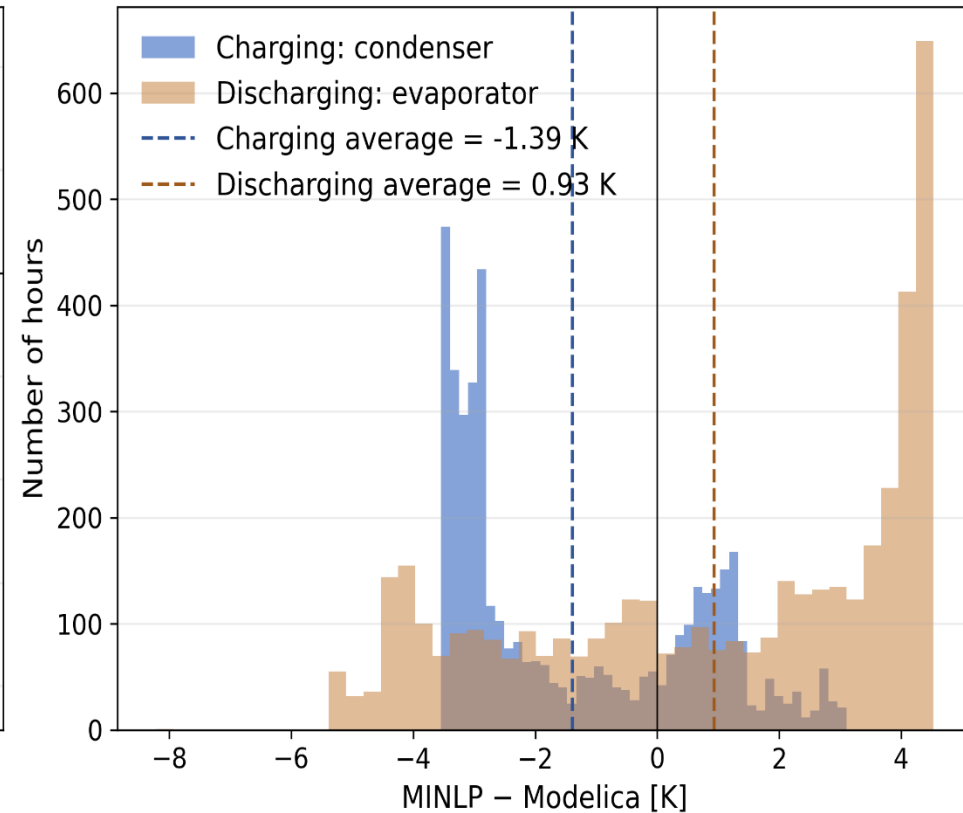
Segment-wise McCormick envelope (since $Q_{LB} = 0$)

$$\begin{aligned} z_{hp,t,s} &\geq T_{evap,s}^{L} Q_{hp,t,s}, & \forall t, s. \\ z_{hp,t,s} &\geq \bar{Q}_{hp,t} T_{evap,t,s} + T_{evap,s}^{U} Q_{hp,t,s} - \bar{Q}_{hp,t} T_{evap,s}^{U} y_{hp,t,s}, & \forall t, s. \\ z_{hp,t,s} &\leq \bar{Q}_{hp,t} T_{evap,t,s} + T_{evap,s}^{L} Q_{hp,t,s} - \bar{Q}_{hp,t} T_{evap,s}^{L} y_{hp,t,s}, & \forall t, s. \\ z_{hp,t,s} &\leq T_{evap,s}^{U} Q_{hp,t,s}, & \forall t, s. \end{aligned}$$

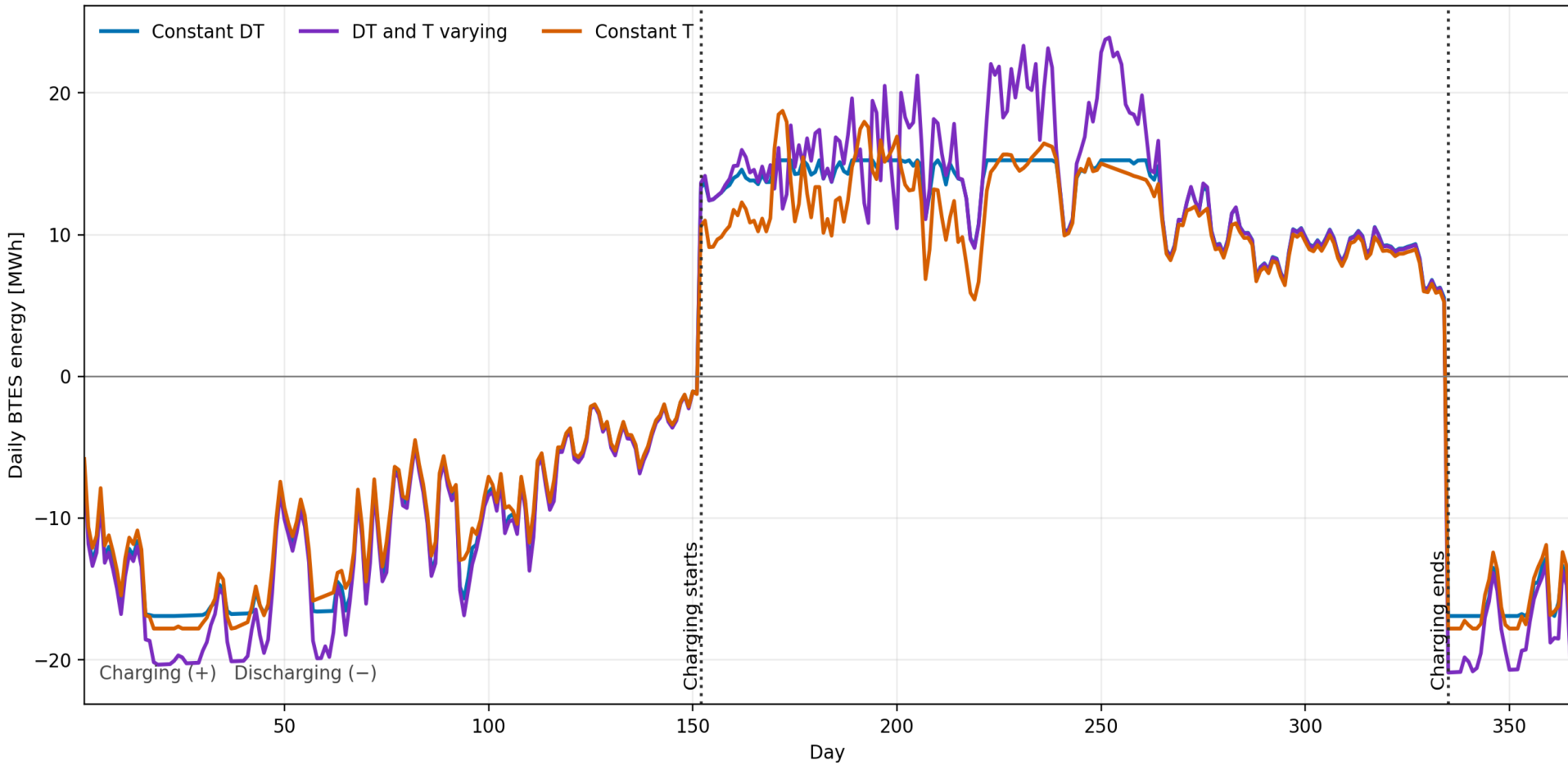
Error distribution by operating period



Hourly error histogram



Quantity	Charging: condenser	Discharging: evaporator
Mean MINLP temperature [°C]	29.926	10.868
Mean Modelica temperature [°C]	31.315	9.936
Mean bias error [K]	-1.389	0.932
Median error [K]	-2.307	1.796
Mean absolute error [K]	2.141	2.885
Root-mean-square error [K]	2.406	3.217
Maximum absolute error [K]	3.543	8.213



Results - Mathematical Formulations

Technology	MINLP	3seg. MILP	Heuristic
Heat Pump (kW)	1139	683	1341
Boiler (kW)	1738	1589	2236
Cooling Tower (kW)	723	624	-
BTES	4	5	4
Volume (m ³)	404,100	606,100	404,100
Annual Cost (kCHF)	995	1056	-
Bilinear Cnstrains	17520	-	-
Binary Variables	26280	26286	-
Run time (s)	118	306	2

Linearization substantially increases the computational effort of the optimization problem

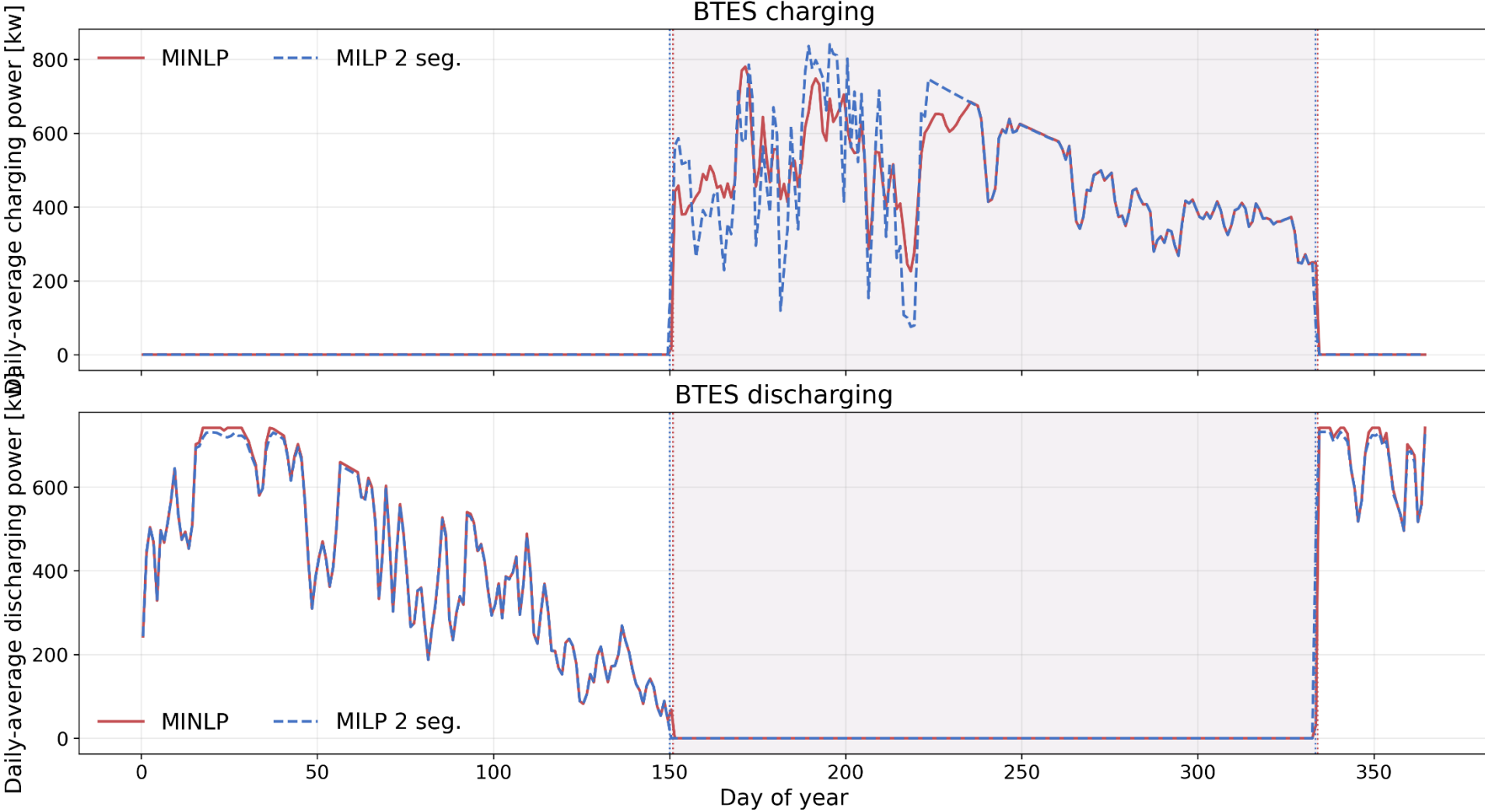
Interpolation of BTES options and 2 seg. MILP

Parameter	Option 4'	Option 4''
Diameter	84.0	87.8
Volume ($10^3 m^3$)	471.4	538.8
Number of boreholes	392	429

The relaxation of the McCormick envelope in this case lies closer to the optimal solution because of the different relaxation bounds

Technology	MINLP	Interpolated MILP	2 seg. MILP
Heat Pump (kW)	1139	1220	1084
Boiler (kW)	1738	1738	1589
Cooling Tower (kW)	723	610	725
BTES	4	4''	4
Volume (m^3)	404,100	471,400	404,100
Annual Cost (kCHF)	995	999	987
Run time (s)	118	282	183

Two-segment linearization: MINLP and MILP operation



Results - MILP simplifications/sizes

Table 7: Comparison of the formulations for the constant condenser and evaporator temperature strategy.

Quantity	MINLP	3 seg.	1 seg.	Fixed period	Fixed period/sizes	Fixed temperatures
McCormick temperature segments	3	3	1	1	1	0
Continuous variables	332 895	332 895	262 812	262 812	262 812	210 250
Binary variables	26 280	26 286	26 283	3	3	1
Total runtime [s]	118.84	292.77	169.38	90.03	106.99	11.14
Speedup relative to MINLP	1.00	0.41	0.70	1.32	1.11	10.67
BTES heat pump [kW]	1139.96	683.49	956.72	951.33	951.33	688.43
Boiler [kW]	1738.70	1589.97	1589.97	1738.70	1738.70	1738.70
BTES chiller [kW]	1247.12	1247.12	1247.12	1247.12	1247.12	1247.12
Cooling tower [kW]	723.76	624.88	725.74	723.66	723.66	624.88
BTES option	4	5	4	4	4	5
BTES depth [m]	80.1	91.7	80.1	80.1	80.1	91.7
CAPEX [CHF/a]	199 122	220 051	190 332	191 127	191 127	221 293
OPEX [CHF/a]	796 169	836 062	793 543	795 398	795 398	836 007
Total annual cost [CHF/a]	995 291	1 056 113	983 874	986 526	986 526	1 057 300
Cost difference from MINLP [%]	0.00	6.11	-1.15	-0.88	-0.88	6.23
Temperature RMSE [K]	0.000	7.064	0.200	0.248	0.252	6.957

Technology	MINLP	1 seg. MILP	MILP-MINLP [%]
BTES Heat Pump (kW)	297	326	10.11
Boiler (kW)	875	866	1.04
Cooling Tower (kW)	829	830	0.06
ASHP (kW)	1064	1043	1.97
Air Chiller (kW)	356	356	0.00
BTES	1	1	-
Volume (m ³)	179,600	179,600	-
Annual Cost (kCHF)	825	802	3.44
Run time (s)	366	253	30.91

- Provides a simple, non-optimization baseline for sizing the borehole thermal energy storage (BTES).
- Stores the smaller of the available summer waste heat and winter heating demand.
- Selects the smallest BTES design capable of storing the required energy.
- Simulates daily charging, discharging, thermal losses, and BTES temperature.

- Available waste heat: $E_{WH} = \sum Q_{cool} \left(1 + \frac{1}{COP_{chiller}} \right)$
- BTES-side heating demand: $E_{dem,BTES} = E_{heat} \left(1 - \frac{1}{COP_{HP}} \right)$
- Required storage energy: $E_{BTES} = \min(E_{WH}, E_{dem,BTES})$
- Number of boreholes: $N_{BH} = \left\lceil \frac{E_{BTES}}{E_{capacity,BH}} \right\rceil$
- Daily BTES temperature: $T_{t+1} = T_t + \frac{Q_{charge,t} - Q_{discharge,t} - Q_{loss,t}}{C_{BTES}}$
- Useful heat delivered: $E_{useful} = \frac{E_{BTES} - E_{loss}}{1 - \frac{1}{COP_{HP}}}$

Decision Variables

Technology	Decision variable	Code name	Meaning	Technology	Decision variable	Code name	Meaning
BTES	T_t^{BTES}	T_btes[t]	BTES temperature	BTES HP	Q_t^{HP}	Q_hp_btes[t]	Heating output
	$Q_t^{ch,chiller}$	Q_charge_chiller[t]	Heat charged from chiller		P_t^{HP}	P_hp_btes[t]	Electrical consumption
	Q_t^{dis}	Q_discharge[t]	Heat extracted from BTES		$COP_t^{HP,-1}$	invCOP_hp_btes[t]	Inverse COP
	ΔT_t^{HP}	DT_hp[t]	HP/BTES temperature difference		S^{HP}	S_hp_btes	HP capacity
	ΔT_t^{CH}	DT_ch[t]	Chiller/BTES temperature difference		$T_{evap,t}$	T_evap_hp[t]	Evaporator temperature
	δ_t^{start}	btes_charge_period_start[t]	charging-period start	BTES Chiller	Q_t^{CH}	Q_ch_btes[t]	Cooling output
	δ_t^{end}	btes_charge_period_end[t]	charging-period end		P_t^{CH}	P_ch_btes[t]	Electrical consumption
	δ_t^{charge}	btes_charging_period[t]	charging-period state		S^{CH}	S_ch_btes	Chiller capacity
	δ_i^{BTES}	delta_btes[i]	BTES design-option		$T_{cond,t}$	T_cond_chiller[t]	Condenser temperature
Electricity	P_t^{grid}	P_import[t]	Electricity imported	Cooling Tower	Q_t^{CT}	Q_cooling_tower[t]	Rejected heat
Natural Gas	G_t^{grid}	G_import[t]	Natural-gas import		S^{CT}	S_cooling_tower	Cooling-tower capacity
				Boiler	Q_t^B	Q_boiler[t]	Boiler heat output
					G_t^B	G_boiler[t]	Gas input
					S^B	S_boiler	Boiler capacity